11.3 Solving Problems Involving Triangles Inscribed in Circles

Lesson Introduction

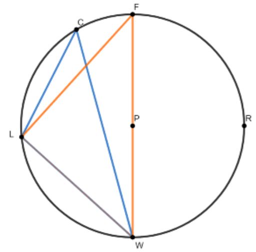
Benchmark Focus	MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.	Learning Targets	• I can solve mathematical problems involving triangles inscribed in a circle.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How do you solve problems involving triangles inscribed in a circle? • What is the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles?
Lesson Narrative	In this lesson, students will apply their knowledge of triangles inscribed in a circle to solve mathematical problems. Students initially explore the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles. Next, students practice solving problems requiring them to use the relationship to find angle and arc measurements.

Component Summary				
	11.3.1 Warm-Up (~5 min.)	Solve a number puzzle *(SE p. 192)*	11.3.3 Guided Practice (10-15 min.)	Guided Practice solving problems using the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles *(SE p. 193)*
	11.3.2 Exploration (15 min.)	Explore the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles *(SE p. 192)*	11.3.4 Your Turn! (10 min.)	Independent practice solving problems using the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles *(SE p. 195)*
	11.3.5 Wrap-Up (~5 Min)	Students summarize their learning *(SE p. 196)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Glossary: • Inscribed Angle • Intercepted Arc	• Rulers • Patty Paper • (Optional) Computers	• Prior to the lesson, work through 11.3.2 Exploration, 11.3.3 Guided Practice, and 11.3.4 Your Turn! to ensure that you are familiar with the steps students will take and are prepared to support them as they work.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not understand the relationship between the inscribed angle and the intercepted arc.
Students may not understand that inscribed angles in different triangles can have the same measurement if they share the same intercepted arc.
Students may not understand that the angle or arc measurements are represented by evaluated algebraic expression, and may provide the value of the variable.</td><td>Consider creating an anchor chart with the relationship between the inscribed angle and the intercepted arc during the debrief of 11.3.2 Exploration.
Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the inscribed angle and in the intercepted arc.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns
In 11.3.3 Guided Practice, students have an opportunity to solve problems using the relationship between the inscribed angle and intercepted arc in triangles inscribed in circles. Prompt students to work with a partner and take turns explaining how they solved the problem and sharing if they agree or disagree and why.</td><td>MA.K12.MTR.3.1: Mathematical Fluency
Students first explore the relationship between the inscribed angle and the intercepted arc in triangles inscribed in circles during 11.3.2 Exploration. Next, they practice solving a variety of problems, first guided in 11.3.3 Guided Practice, and then independently in 11.3.4 Your Turn!. This scaffolded practice will help students build knowledge, skills, and confidence with the lesson learning target.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson provides multiple entry points for visual learners by providing diagrams to represent each question. Auditory learners will have opportunities to discuss with peers the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles. The constructions in 11.3.2 Exploration provides a hands-on approach for kinesthetic learners.

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<tr><td colspan="3">Lesson Notes:</td></tr>
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11.3.1 Warm-Up: Patterns

***Component Overview**: Students investigate a number pattern displayed on rocks arranged in a triangle. Students first pose questions about the pattern and then describe the pattern. Solving non-routine problems, such as this puzzle, can help students build strategic competence.*

11.3.1 Warm-Up: Patterns

1. What is the first question that comes to your mind?
   Answers will vary. What happened to the numbers in the middle? Where is the 5?

2. Describe the pattern used in the image.
   The left-hand side of the rocks is being multiplied by two each time. The right-hand side of the rocks is being multiplied by three each time.

Notice and Wonder
Prompt students to reflect on what they notice and wonder about the pattern. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and high ceiling, as students can be challenged to think of multiple noticings and wonderings.

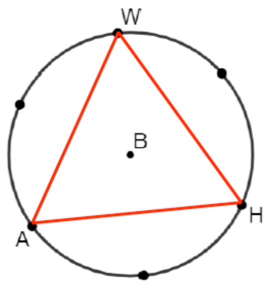
11.3.2 Exploration: Triangles Inscribed in Circles

Component Overview: *Students explore the relationship between inscribed angles and intercepted arcs in an equilateral triangle inscribed in a circle.*

11.3.2 Exploration: Triangles Inscribed in Circles

Work with your partner to complete the following.

Elijah is constructing a triangle inscribed in Circle B and stopped after he created six points on Circle B that are the same distance from each other.



1. Using a straightedge, draw a line from point A to point H.
   Answer on image above

2. Using a straightedge, draw a line from point H to point W.
   Answer on image above

3. In relation to the circle, what type of angle is $\angle AHW$?
   An inscribed angle

4. Explain the relationship between $m\angle AHW$ and $m\widehat{WA}$.
   $m\angle AHW$ is $\frac{1}{2}\left(m\widehat{WA}\right)$ because an inscribed angle is half the measure of its intercepted arc.

This Exploration is designed to allow students to engage with it with a partner, rather than being guided through by the teacher. Observe and listen as students work with their partners. There will be an opportunity to summarize and synthesize what is discovered in the next component.

Consider creating an anchor chart with the relationship between the inscribed angle and the intercepted arc.

Encourage students to discuss their explanation for question 4 with their partner to provide an auditory entry point. The use of a ruler, straightedge, and/or patty paper provide an entry point for kinesthetic learners.

11.3.2 Exploration: Triangles Inscribed in Circles (continued)

5. Using a straightedge, draw a line from point A to point W.
   Answer on image above

6. Use a ruler or patty paper to verify that ΔAHW is an equilateral triangle.
   Answers will vary.

7. What is the $m\angle AHW$?
   $60°$

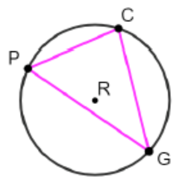
8. What is the $m\widehat{WA}$?
   $120°$

11.3.3 Guided Practice: Triangles Inscribed in Circles

Component Overview: *Students engage in Guided Practice by solving problems using the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles.*

11.3.3 Guided Practice: Triangles Inscribed in Circles

1. Circle R is shown. Points P, C, and G are on circle R and create ΔPCG. $m\,\widehat{PG} = 162°$, $m\angle CPG = 58°$, and $m\angle PCG = (7x - 17)°$.

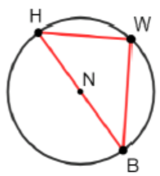


   a. What is the $m\angle PCG$?
   $81°$

   b. Find the value of x.
   $x = 14$

   c. What is the $m\angle PGC$?
   $41°$

2. Circle N has points H, W, and B on the circle and diameter $\overline{HB}$. ΔHWB is inscribed in circle N. $m\,\widehat{BW} = 102°$ and $m\angle HBW = (4x - 10)°$.



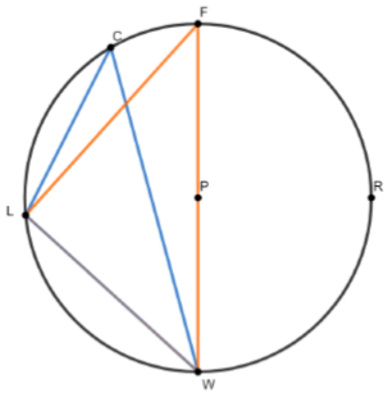
   a. What is the $m\angle HBW$?
   $39°$

   b. Find the value of x.
   $x = 12.25$

Use observational data from 11.3.2 Exploration to determine how much guidance is needed to bring together key ideas in this Guided Practice. If necessary, consider modeling question 1 or pulling a small group as they work through that problem to address common misconceptions.

Take Turns
Students have an opportunity to solve problems using the relationship between the inscribed angle and intercepted arc in triangles inscribed in circles. Prompt students to work with a partner and take turns explaining how they solved the problem and sharing if they agree or disagree and why.

3. ΔCLW and ΔFLW are inscribed in Circle P, where $\overline{FW}$ is the diameter of Circle P. $m\widehat{FC} = (2x - 6)°$, $m\widehat{LC} = (4x - 6)°$, and $m\widehat{LW} = (3x + 30)°$.



Complete the table.

Angle Measures	Angle Measures
$m\angle LCW = 42°$	$m\angle LFW = 42°$
$m\angle CWL = 33°$	$m\angle FWC = 15°$
$m\angle FLC = 15°$	$m\angle FLW = 90°$

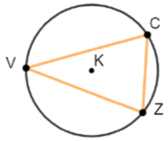
Consider prompting students to separate this circle into two separate circles, with one triangle inscribed on each circle. Students can trace the circle and inscribed triangles on patty paper.

11.3.4 Your Turn!

Component Overview: *Students practice solving problems independently using the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles. Circulate while students work, to support struggling students and determine which students you will ask to share their strategies during a debrief discussion.*

11.3.4 Your Turn!

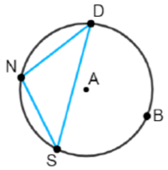
1. Circle K has points V, C, and Z on the circle, and ΔVCZ is inscribed in circle K. $m\widehat{VC} = (9x + 23)°$, $m\angle CVZ = 36°$, and $m\angle VZC = (7x - 21)°$.



   a. What is the value of x?
   $x = 13$

   b. What is the $m\angle VCZ$?
   $74°$

2. Circle A has points N, D, B, and S on the circle, and ΔSDN is inscribed in Circle A. $m\widehat{ND} = (7x - 7)°$, $m\widehat{DBS} = (23y - 5)°$, $m\angle NSD = (5x - 23)°$, and $m\angle DNS = (8y + 29)°$.



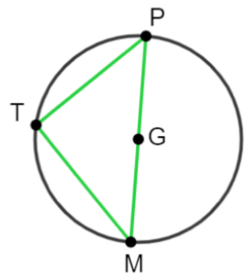
   Complete the table.

Angle Measures	Arc Measures
$m\angle NSD = 42°$	$m\widehat{ND} = 84°$
$m\angle SND = 101°$	$m\widehat{DBS} = 202°$
$m\angle NDS = 37°$	$m\widehat{NS} = 74$

Students first explore the relationship between the inscribed angle and the intercepted arc in triangles inscribed in circles during 11.3.2 Exploration. Next, they practice solving a variety of problems, first guided in 11.3.3 Guided Practice, and then independently in 11.3.4 Your Turn!. This scaffolded practice will help students build knowledge, skills, and confidence with the lesson learning target. Debrief each component with a discussion to check for student understanding.

Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the inscribed angle and in the intercepted arc.

3. Circle G has diameter $\overline{MP}$ and ΔMTP inscribed in the circle. $m\angle TMP = (6x - 8)°$ and $m\angle TPM = (4x + 13)°$.



a. Find the value of x.
$x = 8.5$

b. What is the $m\angle TMP$?
$43°$

c. What is the $m\angle TPM$?
$47°$

- *How did you determine the inscribed angle measures and the intercepted arc measure?*
- *Why is it important that the problem stated that the diameter of circle G was MP?*
- *How could you verify that your angle measures are correct?*

11.3.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

11.3.5 Wrap-Up: Cool Down

1. Write a summary of what you learned about solving problems involving triangles inscribed in circles.

   Answers will vary.

Consider providing students with sentence stems or answer options to choose from. These strategies will help students compose written responses to the prompt.